

Project Architecture

Project architecture is the high-level structure of a software project that defines how components, modules, services, data, and infrastructure interact with each other.

Typical Project Architecture

```
Project
├── client/
├── server/
├── database/
├── shared/
├── tests/
├── docs/
├── deployment/
├── .env
├── README.md
└── package.json
```

Layered Architecture

```
Presentation Layer
↓
API Layer
↓
Business Layer
↓
Data Access Layer
↓
Database
```

Key Principles

- Separation of Concerns
- Scalability
- Maintainability
- Reusability
- Security
- Testability

Example for a Web Application

```
Frontend
↓
API Gateway
↓
Backend Services
- User Service
- Product Service
- Order Service
↓
Database
- PostgreSQL
- Redis Cache
```